1. Purpose & Overview

These instructions define the mandatory standardized settings, protocols, and submission requirements for the Federated Learning (FL) based experiments. Every group must follow these guidelines precisely so that results across different research categories can be meaningfully compared, aggregated, and discussed collectively at the end of the course.

Each group will:

- Read and survey the 2 assigned research papers to each group
- Reproduce or extend the core experiments using the Flower (flwr) framework
- Follow the standardized experimental settings defined in this document
- Submit all deliverables by the stated deadlines

2. Software Environment & Setup

2.1 Mandatory Framework

All experiments must be implemented using Flower (flwr), the standard federated learning framework for this course.

Framework	Flower (flwr) — https://flower.ai
Python Version	Python 3.9 or higher (3.10 recommended)
Deep Learning	PyTorch >= 2.0 preferred.
Seed (Fixed)	random.seed(42) | numpy.seed(42) | torch.manual_seed(42)

Github Repo	Each Group must submit a private repo

3. Experimental Settings

Try keeping clients as 10, 50, 100, and datasets, as MNIST, FMNIST, CIFAR-10, CIFAR-100, Shakespeare, Sent140. You may adapt sometimes if required, in the paper, you may reach out to help.

4. Universal Baseline Configuration

Regardless of category, All groups must implement FedAvg as the baseline algorithm first, using the common settings. Your category-specific experiments then build on top of this baseline.

4.1 Federated Averaging (FedAvg) Baseline

Algorithm	FedAvg (McMahan et al., 2017)
Client Fraction per Round	0.5 (50% of total clients sampled per round)
Local Epochs	5 epochs per communication round
Local Batch Size	32
Optimizer	SGD with momentum = 0.9
Learning Rate	0.01
Loss Function	Cross-Entropy Loss
Weight Initialization	default

4.3 Data Partitioning (Non-IID Default)

For all categories except Category 3 (Data Heterogeneity), use the following default data partitioning strategy so that the baseline is consistent across groups:

Partitioning Method	Use Dirichlet distribution $\alpha \in \{0.01, 0.1, 0.5, 1.0, \text{IID}\}$ and report results.

5. Evaluation Metrics

All groups must report the following metrics in their results. Additional category-specific metrics are listed below.

5.1 Universal Metrics (All Groups)

- Global Test Accuracy — evaluated every round on a held-out global test set
- Global Test Loss — cross-entropy loss on the global test set
- Convergence Round — the round at which accuracy first exceeds 80% (or domain-appropriate threshold)
- Communication Cost — total MB transmitted (model size × clients × rounds)

5.2 Category-Specific Metrics

Please consider adding the metrics mentioned in the paper.

Cat. 1 – Privacy & Inference	Attack success rate, reconstruction MSE, membership inference AUC
Cat. 2 – Poisoning & Adversarial	Backdoor attack success rate, clean accuracy under attack, defense recovery rate
Cat. 3 – Data Heterogeneity	Per-client accuracy distribution, variance across clients, fairness gap
Cat. 4 – Communication Efficiency	Compression ratio, accuracy-communication tradeoff curve
Cat. 5 – Personalization	Per-client personalized accuracy, global vs. local accuracy tradeoff
Cat. 6 – Fairness & Bias	Accuracy across demographic groups, fairness metrics (equalized odds, DP)
Cat. 7 – System Heterogeneity	Round completion time, straggler ratio, accuracy under partial participation
Cat. 8 – Cross-Silo / Decentralized	Gossip convergence vs. centralized FL, silo-level accuracy
Cat. 9 – FL with LLMs	Perplexity, BLEU/ROUGE score, fine-tuning accuracy, LoRA parameter efficiency

Cat. 10 – Continual FL	Catastrophic forgetting ratio, backward transfer, task-specific accuracy

6. Mandatory Reporting Format

All experiments must be reported using a standardized format so results can be directly compared across groups. Failing to use this format will result in deduction of marks.

6.1 Figures (Mandatory Plots)

- Global accuracy vs. communication rounds (for all experiments on one plot)
- Global loss vs. communication rounds
- Category-specific metric vs. rounds (e.g., attack success rate for Cat. 2)
- IID vs. Non-IID comparison plot
- Baseline (FedAvg) vs. your proposed/studied method — side-by-side bars or line plot

Figure requirements:

  - Resolution: 300 DPI minimum
  - Format: PNG or PDF
  - Font size in figures: minimum 11pt
  - Each figure must have a descriptive caption and be numbered (Figure 1, Figure 2, …)

6.2 Results Table (Mandatory)

Include a summary table in your report comparing all experimental configurations:

- Columns: Method, Dataset, #Clients, #Rounds, Test Accuracy (%), Convergence Round, Comm. Cost (MB), Category Metric
- One row per experiment configuration
- Highlight the best result in bold

7. Submission Requirements

All deliverables must be submitted by the deadline.

Deliverable	Description	Format
Survey Report	Literature review of assigned papers + related work summary (containing all the images)	Overleaf 8-9 pages in 2 col format. Min 7 pages 2 col. Don’t make figures excessive large to increase page count.

Deliverable	Description	Format
		Link—use this template strictly: https://www.overleaf.com/latex/templates/ieee-journal-paper-template/jbbbdkztwxrd
Code Repository	All experiment code, configs, and README on GitHub	Pvt GitHub repo link Only one member needs to send invite.
Youtube Video	A complete workflow on how to execute the project	Save in the github readme.md
Plag and AI report	Submit your final overleaf pdf to al@iitp.ac.in and plag <=14 % and AI content should show * via Turnitin. Ai content <5 is fine	

7.1 GitHub Repository Structure

Your repository should follow this exact structure:

- README.md — setup instructions, group info, category, paper titles, youtube video
- configs/ — all YAML config files for each experiment
- src/ — all source code (client.py, server.py, model.py, utils.py, data.py)
- results/ — saved metrics as .csv and plots as .png/.pdf
- report/ — final report PDF
- requirements.txt — all dependencies with pinned versions